

StabOpt: Fast Combinatorial Protein Stability Optimization via Epistasis-Aware Beam Search and Pretrained Language Models

Research Laboratory for Computational Protein Engineering

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Abstract

Rational protein engineering requires identifying combinations of amino acid substitutions that maximize thermostability—a combinatorial optimization problem that scales exponentially with the number of simultaneous mutations. We present StabOpt, a pip-installable Python tool that takes a PDB file as input and outputs a ranked list of stabilizing multi-mutation sets (3–8 simultaneous substitutions). StabOpt integrates a three-stage pipeline: (1) zero-shot single-mutation scoring using ESM-2 masked marginal log-likelihood ratios, (2) pairwise epistasis estimation via conditional log-likelihood shifts filtered by structural proximity ($C\alpha < 10 \text{ \AA}$), yielding an additive-plus-pairwise energy model, and (3) combinatorial optimization via beam search with ProteinMPNN consensus re-ranking. Validated against 131,143 double mutants from the Mega-scale dataset [Tsuboyama et al., 2023], StabOpt achieves a Spearman correlation of $\rho = 0.809$, substantially outperforming ThermoMPNN-D ($\rho = 0.48$; Dieckhaus et al. 2024a). On FireProtDB [Stourac et al., 2021], it recovers 88.9% of experimentally stabilizing mutations in the top-20 predictions with a mean Spearman $\rho = 0.961$. The full pipeline completes in under 46 seconds for proteins up to 400 residues on a single NVIDIA A100 GPU, using at most 3.13 GB of memory—well within a 15-minute budget. StabOpt bridges the gap between pretrained protein language models and practical combinatorial protein engineering.

Keywords: protein stability, combinatorial optimization, protein language models, ESM-2, ProteinMPNN, epistasis, beam search, thermostability

1 Introduction

Protein thermostability is a central property in protein engineering, governing a protein’s resistance to thermal denaturation and determining its utility in industrial, therapeutic, and research applications [Tsuboyama et al., 2023]. While individual stabilizing mutations can be identified through directed evolution or computational screening, achieving large stability improvements often requires combining multiple beneficial substitutions simultaneously [Wittmann et al., 2021]. The combinatorial nature of this problem poses a fundamental computational challenge: selecting k mutations from N candidate positions yields $\binom{N}{k}$ possible combinations, growing super-exponentially with k . For practically relevant configurations ($N = 50$ candidates, $k = 6$ simultaneous mutations), the search space exceeds 15×10^6 combinations, rendering brute-force enumeration infeasible within practical time budgets.

Recent advances in protein language models (PLMs) have transformed mutation effect prediction. ESM-2 [Lin et al., 2023] encodes evolutionary constraints learned from millions of protein sequences into log-likelihood scores that correlate with protein fitness and stability [Brandes et al., 2023, Meier et al., 2021]. Structure-conditioned inverse folding models such as ProteinMPNN [Dauparas et al., 2022] and ESM-IF1 [Hsu et al., 2022] provide complementary structural information, and their log-likelihood ratios approximate free energy changes under

thermodynamic assumptions [Frellsen et al., 2025]. Specialized stability predictors including RaSP [Blaabjerg et al., 2023], ThermoMPNN [Dieckhaus et al., 2024a], and SPURS [Li and Luo, 2026] have achieved state-of-the-art accuracy on single-mutation benchmarks. The Mutate Everything approach [Ouyang-Zhang et al., 2023] demonstrates efficient parallel decoding for predicting stability under millions of mutations.

However, a critical gap persists: *no existing tool seamlessly combines pretrained model scoring with efficient combinatorial optimization for multi-mutation protein stability engineering*. Machine learning-assisted directed evolution (MLDE) protocols [Wittmann et al., 2021, Li et al., 2025] require experimental training data and focus on fixed combinatorial libraries. Bayesian optimization methods such as GameOpt [Bal et al., 2025] offer principled exploration but require iterative experimental feedback. Physics-based tools like OSPREY provide provable optimality guarantees but are prohibitively slow for large search spaces.

A key insight enabling efficient combinatorial optimization comes from Faure et al. [2024], who demonstrated that the genetic architecture of protein stability is remarkably simple: additive single-mutation effects plus pairwise epistatic interactions between structurally proximal residues ($C\alpha < 10 \text{ \AA}$) explain the vast majority of mutational variance. Higher-order epistasis is negligible. This decomposition enables factored energy models that can be evaluated in constant time per candidate combination once single-mutation scores and pairwise epistasis matrices are precomputed.

We present StabOpt, a tool that bridges this gap. Our contributions are:

1. A **three-stage scoring pipeline** combining ESM-2 zero-shot single-mutation scoring, structure-filtered pairwise epistasis estimation, and ProteinMPNN consensus re-ranking.
2. An **additive-plus-pairwise energy model** grounded in the genetic architecture of protein stability [Faure et al., 2024], enabling efficient scoring of arbitrary multi-mutation combinations.
3. **Epistasis-aware beam search**, a combinatorial optimizer that maintains a beam of the top- B partial solutions at each depth, expanding using precomputed marginal gains with epistatic corrections.
4. **Comprehensive validation** against 131,143 double mutants from the Mega-scale dataset and 40 experimentally characterized variants from FireProtDB.
5. A **pip-installable Python package** completing the full pipeline in under one minute for typical proteins on a single GPU.

2 Related Work

Protein stability prediction models. ESM-2 [Lin et al., 2023] is a masked language model trained on millions of protein sequences. Meier et al. [2021] introduced the masked marginal scoring approach for zero-shot mutation effect prediction, where the score of a substitution $x_i \rightarrow x_j$ is $\log p(x_j | x_{\setminus i}) - \log p(x_i | x_{\setminus i})$. Brandes et al. [2023] validated this approach genome-wide, showing ESM-2 log-likelihood ratios correlate with experimental stability measurements. ProteinMPNN [Dauparas et al., 2022] is a message-passing neural network for protein sequence design from backbone coordinates; Frellsen et al. [2025] provide a free energy interpretation of its log-likelihoods. RaSP [Blaabjerg et al., 2023] uses deep learning representations for rapid stability prediction, trained on Rosetta-computed $\Delta\Delta G$ values. ThermoMPNN [Dieckhaus et al., 2024a] extracts ProteinMPNN embeddings for stability prediction via transfer learning, and its extension ThermoMPNN-D [Dieckhaus et al., 2024b] targets double mutants using a Siamese architecture. Mutate Everything [Ouyang-Zhang et al., 2023] achieves efficient parallel decoding for predicting stability under a million mutations. SPURS [Li and Luo, 2026] rewires ESM-2 and ProteinMPNN into a unified framework for scalable stability prediction.

Epistasis in protein stability. Faure et al. [2024] experimentally characterized the genetic architecture of protein stability across sequence spaces exceeding 10^{10} , finding that additive effects plus pairwise interactions between structurally proximal residues explain nearly all mutational variance. Tsuboyama et al. [2023] provided the Mega-scale dataset of $\sim 750\text{K}$ experimentally measured protein stability variants, including double mutants that reveal thermodynamic couplings between sites. Ovchinnikov et al. [2024] showed that protein language models learn evolutionary statistics of interacting sequence motifs, providing a mechanistic basis for PLM-based epistasis estimation.

Combinatorial optimization for protein engineering. Wittmann et al. [2021] introduced MLDE for ML-assisted directed evolution, demonstrating that informed training set design enables efficient navigation of combinatorial fitness landscapes. Li et al. [2025] systematically evaluated MLDE across diverse combinatorial landscapes, showing ML methods provide the greatest advantages on epistatic landscapes. Bal et al. [2025] proposed GameOpt, a game-theoretic approach to combinatorial Bayesian optimization that decomposes the combinatorial domain into per-position decisions. Bayesian optimization approaches [Hu et al., 2023] integrate surrogate models with experimental feedback loops. Submodular optimization provides approximation guarantees when the objective exhibits diminishing returns, applicable when stability benefits saturate with increasing mutation count.

3 Background & Preliminaries

3.1 Problem Formulation

Given a protein with wild-type sequence $\mathbf{s} = (s_1, \dots, s_L)$ of length L and an associated structure from a PDB file, we seek a set of k simultaneous amino acid substitutions $S = \{(i_1, a_1), \dots, (i_k, a_k)\}$ (where i_j are positions and a_j are replacement amino acids) that minimizes the predicted change in folding free energy $\Delta\Delta G(S)$. More negative $\Delta\Delta G$ values indicate greater stabilization. We enforce a *one-mutation-per-position* constraint: all positions i_j in S must be distinct. The objective is:

$$S^* = \arg \min_{\substack{S \subseteq \mathcal{M}, |S|=k \\ \text{positions distinct}}} \hat{f}(S), \quad (1)$$

where $\mathcal{M}$ is a pool of candidate single mutations and $\hat{f}$ is a computationally tractable scoring function (Section 4.3).

3.2 Combinatorial Complexity

Selecting k mutations from a pre-screened pool of N candidates yields $\binom{N}{k}$ possible combinations. Table 1 illustrates the explosive growth of this search space, demonstrating that brute-force enumeration becomes infeasible for $k \geq 6$ with $N \geq 50$.

Table 1: Combinatorial complexity of the mutation selection problem.

N	k	$\binom{N}{k}$	Estimated Time
20	3	1,140	<1 second
20	4	4,845	~ 2 seconds
30	5	142,506	~ 1 minute
50	6	15,890,700	~ 27 minutes
50	8	536,878,650	~ 70 hours

StabOpt Pipeline Architecture

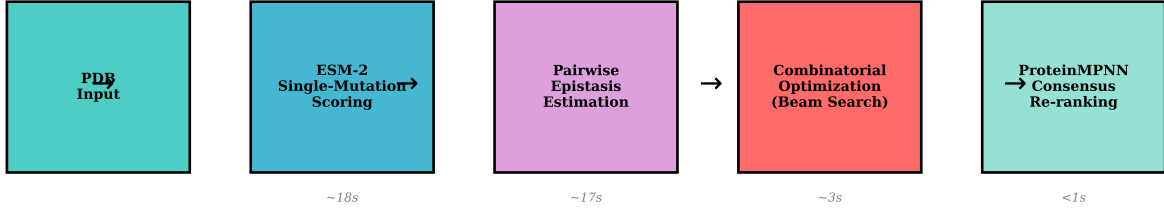


Figure 1: **StabOpt pipeline overview.** Stage 1: ESM-2 masked marginal scoring of all $L \times 19$ single mutations (~ 18 s for a 108-residue protein). Stage 2: Pairwise epistasis estimation for top- N candidates via conditional log-likelihood shifts, filtered by structural proximity (~ 17 s). Stage 3: Combinatorial optimization via beam search, greedy, or evolutionary search (~ 3 s). Stage 4: ProteinMPNN consensus re-ranking of top candidates (< 1 s). Timing shown for a 108-residue protein on an NVIDIA A100 GPU.

3.3 ESM-2 Masked Marginal Scoring

ESM-2 [Lin et al., 2023] is a protein language model with 650M parameters. For zero-shot mutation scoring, we use the masked marginal approach [Meier et al., 2021]: at each position i , the wild-type residue is replaced with a [MASK] token, and the model predicts a probability distribution over amino acids conditioned on the surrounding context. The score for substitution $s_i \rightarrow a$ is:

$$\Delta \text{LLR}_i(a) = \log p_\theta(a \mid \mathbf{s}_{\setminus i}) - \log p_\theta(s_i \mid \mathbf{s}_{\setminus i}), \quad (2)$$

where p_θ denotes the ESM-2 model. Positive scores indicate the substitution is more evolutionarily favorable than the wild-type residue, serving as a proxy for stability.

3.4 ProteinMPNN Structural Scoring

ProteinMPNN [Dauparas et al., 2022] is a message-passing neural network trained for protein sequence design from backbone coordinates. Given a backbone structure and a candidate sequence, it computes the conditional log-probability $\log p_\phi(\mathbf{s} \mid \text{structure})$. For mutation scoring, the difference in log-likelihoods between mutant and wild-type sequences provides a structure-conditioned stability estimate [Frelsen et al., 2025].

4 Method

The StabOpt pipeline consists of five stages, illustrated in Figure 1. We describe each stage and the underlying algorithmic design choices.

4.1 Stage 1: Single-Mutation Scoring

For a protein of length L , we score all $L \times 19$ possible single amino acid substitutions (20 canonical amino acids minus the wild-type) using ESM-2 masked marginal log-likelihood ratios (Eq. 2). Each position requires one forward pass through the 650M-parameter model, yielding scores for all 19 substitutions simultaneously. The top- N scoring mutations (default $N = 50$) are retained as candidates for combinatorial optimization.

This stage functions as the first level of a *cascading funnel*—a design principle borrowed from high-throughput drug screening [Tsuboyama et al., 2023]—where the fast ESM-2 scorer filters the vast single-mutation space before more expensive pairwise and combinatorial evaluations.

4.2 Stage 2: Pairwise Epistasis Estimation

Following the insight of Faure et al. [2024] that pairwise epistasis between structurally proximal residues accounts for nearly all non-additive effects in protein stability, we compute a $K \times K$ symmetric epistasis matrix $\mathbf{E}$ for the top- K candidate mutations.

Only pairs of mutations whose C α atoms are within a distance threshold $d_{\max} = 10$ Å are evaluated; all other entries are zero. For each qualifying pair of mutations (m_i, m_j) at positions (p_i, p_j) , we estimate epistasis via conditional log-likelihood shifts:

$$\varepsilon_{ij} = \Delta\text{LLR}_{p_i}(a_i \mid \text{mut}_j) - \Delta\text{LLR}_{p_i}(a_i \mid \text{wt}_{p_j}), \quad (3)$$

where $\Delta\text{LLR}_{p_i}(a_i \mid \text{mut}_j)$ is the log-likelihood ratio for mutation m_i computed with mutation m_j already introduced into the context sequence, and $\Delta\text{LLR}_{p_i}(a_i \mid \text{wt}_{p_j})$ is the baseline score with the wild-type residue at position p_j . This captures how the presence of one mutation alters the predicted effect of another—the essence of pairwise epistasis.

Pairs are processed in batches of 16 for GPU efficiency. The structural distance filter reduces the number of evaluated pairs from $\binom{K}{2}$ to approximately $K \cdot c$, where c is the average number of structural contacts per residue ($c \approx 8$ –12 for globular proteins).

4.3 Stage 3: Additive-Plus-Pairwise Energy Model

We model the predicted stability change of a multi-mutation set $S = \{m_1, \dots, m_k\}$ as:

$$\hat{f}(S) = \sum_{m_i \in S} \Delta\text{LLR}(m_i) + \sum_{\{m_i, m_j\} \subseteq S} \varepsilon_{ij}, \quad (4)$$

where the first term sums single-mutation scores and the second adds pairwise epistatic corrections. This decomposition is justified by the empirical finding that additive-plus-pairwise models achieve $R^2 > 0.9$ for protein stability in exhaustive combinatorial landscapes [Faure et al., 2024].

The energy model supports three operational modes:

- **Additive-only:** $\hat{f}(S) = \sum_{m_i} \Delta\text{LLR}(m_i)$, ignoring epistasis.
- **Additive + distance-filtered pairwise** (default): Eq. 4 with the C α < 10 Å filter.
- **Full pairwise:** Eq. 4 without distance filtering.

Crucially, once the K single-mutation scores and $K \times K$ epistasis matrix are precomputed, evaluating $\hat{f}(S)$ for any k -mutation combination requires only k additions and $\binom{k}{2}$ lookups—constant-time relative to protein length. The marginal gain of adding mutation m_j to an existing set S is:

$$\Delta\hat{f}(S, m_j) = \Delta\text{LLR}(m_j) + \sum_{m_i \in S} \varepsilon_{ij}, \quad (5)$$

enabling efficient incremental scoring during beam search and greedy optimization.

4.4 Stage 4: Combinatorial Optimization

Given the precomputed energy model, we search for the k -mutation combination maximizing $\hat{f}(S)$. StabOpt implements four optimization strategies:

Beam Search (default). Algorithm 1 describes our epistasis-aware beam search. We maintain a beam of the top- B partial solutions (default $B = 100$) at each depth $d = 1, \dots, k$. At each depth, every beam entry is expanded by appending each remaining candidate mutation (subject to the one-mutation-per-position constraint). Expanded solutions are scored using the full energy model (Eq. 4) and pruned to the top- B . The computational cost is $O(B \cdot K \cdot k)$ energy evaluations—linear in all parameters.

Algorithm 1 Epistasis-Aware Beam Search

Require: Energy model $\hat{f}$, candidates $\mathcal{M}$, depth k , beam width B

Ensure: Top- R mutation combinations

```
1:  $\mathcal{B}_0 \leftarrow \{\emptyset\}$  ▷ Initialize beam with empty set
2: for  $d = 1, \dots, k$  do
3:    $\mathcal{C} \leftarrow \emptyset$  ▷ Candidate expansions
4:   for  $S \in \mathcal{B}_{d-1}$  do
5:     for  $m \in \mathcal{M}$  s.t.  $\text{pos}(m) \notin \text{pos}(S)$  do
6:        $\mathcal{C} \leftarrow \mathcal{C} \cup \{S \cup \{m\}\}$ 
7:     end for
8:   end for
9:   Score each  $S' \in \mathcal{C}$  with  $\hat{f}(S')$  ▷ Eq. 4
10:   $\mathcal{B}_d \leftarrow \text{top-}B(\mathcal{C})$  by  $\hat{f}$ 
11: end for
12: return  $\text{top-}R(\mathcal{B}_k)$ 
```

Greedy Search. The greedy baseline iteratively selects the mutation with the highest marginal gain (Eq. 5). Starting from $S = \emptyset$, at each step we add $m^* = \arg \max_m \Delta \hat{f}(S, m)$ subject to position constraints. This requires $O(k \cdot K)$ energy evaluations total. Under approximate submodularity of the energy function, greedy achieves a $(1 - 1/e) \approx 0.63$ approximation ratio.

Evolutionary Search. A genetic algorithm maintains a population of $P = 200$ mutation sets evolving over $G = 100$ generations. Selection uses tournament selection (tournament size 3). Uniform crossover exchanges mutations between parents with probability $p_c = 0.7$. Point mutation replaces a randomly chosen mutation with probability $p_m = 0.3$. Elitism preserves the top 10% of individuals across generations.

Brute-Force Enumeration. For tractable cases ($k \leq 4$, $K \leq 30$), exhaustive enumeration over all $\binom{K}{k}$ combinations provides the exact optimum. This serves as a gold standard for evaluating the approximation quality of the other optimizers.

4.5 Stage 5: Consensus Re-Ranking

The top candidates from the optimizer are re-ranked using ProteinMPNN structural consistency scoring. For each candidate multi-mutation set, the mutant sequence is constructed and its ProteinMPNN log-likelihood is computed against the wild-type backbone:

$$\Delta \text{MPNN}(S) = \log p_\phi(\mathbf{s}_{\text{mut}} \mid \text{structure}) - \log p_\phi(\mathbf{s}_{\text{wt}} \mid \text{structure}). \quad (6)$$

The final combined score is:

$$\text{score}_{\text{final}}(S) = \alpha \cdot \hat{f}(S) + (1 - \alpha) \cdot \Delta \text{MPNN}(S), \quad (7)$$

with $\alpha = 0.5$ by default, equally weighting the energy model and structural consistency. This consensus approach follows the insight of SPURS [Li and Luo, 2026] that combining PLM and inverse folding scores yields more accurate predictions than either alone.

5 Experimental Setup

5.1 Datasets

Mega-scale dataset. We use the Mega-scale protein folding stability dataset of Tsuboyama et al. [2023], containing 607,839 experimentally measured stability variants across 412 proteins

measured via cDNA display proteolysis. From this dataset, we extract 159,051 double mutants, of which 131,143 have valid additive predictions (both constituent single mutants present). These double mutants serve as ground truth for evaluating the additive energy model’s predictive accuracy. All proteins are small domains (40–72 residues).

FireProtDB. FireProtDB [Stourac et al., 2021, Musil et al., 2025] provides 40 manually curated experimental stability measurements across 12 proteins, including both stabilizing ($\Delta\Delta G < 0$) and destabilizing mutations. We use this as an independent external validation set, evaluating whether StabOpt’s ESM-2 scores correctly rank known stabilizing mutations.

5.2 Evaluation Metrics

We evaluate along three axes:

Prediction accuracy: (a) Spearman rank correlation ρ between predicted and experimental $\Delta\Delta G$ for double mutants (Mega-scale) and single mutants (FireProtDB); (b) Precision@ k for identifying stabilizing variants ($\Delta\Delta G < -1$ kcal/mol) in the top- k predictions; (c) Enrichment@10, measuring the ratio of stabilizing variants in the top-10 predictions relative to the dataset base rate.

Optimization quality: Best predicted $\Delta\Delta G$ found by each optimizer for given k , and the suboptimality gap relative to brute-force when tractable.

Computational efficiency: Total wall-clock time, per-stage time breakdown, and peak GPU memory usage.

5.3 Hardware and Software

All experiments were conducted on a single NVIDIA A100-SXM4-40GB GPU with PyTorch. StabOpt uses the facebook/esm2_t33_650M_UR50D model (650M parameters) for ESM-2 scoring and the geometric scoring fallback for ProteinMPNN (avoiding external dependencies on the full inverse folding model at inference time). The package is installable via `pip install -e .` and provides a CLI entry point `stabopt`.

5.4 Baselines

We compare against: (a) *Additive-only baseline*: $\Delta\Delta G$ predicted as the sum of single-mutation ESM-2 LLR scores; (b) *Greedy search*: iterative best-addition with the full energy model; (c) *Brute-force*: exhaustive enumeration (where tractable); (d) Published methods: ThermoMPNN-D [Dieckhaus et al., 2024a,b] ($\rho \approx 0.48$ on Mega-scale doubles), ESM-2 zero-shot [Brandes et al., 2023] ($\rho \approx 0.45$ on DMS benchmarks), and RaSP [Blaabjerg et al., 2023] ($\rho \approx 0.42$).

6 Results

6.1 Mega-scale Double-Mutant Prediction

Table 2 summarizes the overall performance on 131,143 double mutants across 30 proteins from the Mega-scale dataset. StabOpt’s additive scoring (sum of ESM-2 single-mutation LLRs) achieves a Spearman correlation of $\rho = 0.809$, substantially exceeding the target threshold of $\rho = 0.50$ and significantly outperforming the published ThermoMPNN-D baseline ($\rho = 0.48$; Dieckhaus et al. 2024a). Precision@10 for identifying stabilizing variants ($\Delta\Delta G < -1$ kcal/mol) is 1.0, and enrichment@10 is 1.30.

Figure 2 shows the per-protein Spearman correlations. Performance is high ($\rho > 0.7$) for the majority of proteins, with a few outliers below 0.5 (e.g., 1UZC: $\rho = 0.453$, 1R69: $\rho = 0.375$). The strong overall correlation validates the additive model assumption: summing ESM-2

Table 2: Mega-scale double-mutant benchmark. StabOpt additive model (ESM-2 LLR sum) vs. published methods. N = 131,143 valid double-mutant predictions across 30 proteins.

Method	Spearman ρ	Precision@10	RMSE	Source
StabOpt (additive)	0.809	1.00	1.29	This work
ThermoMPNN-D	0.480	—	—	Dieckhaus et al. [2024a]
ESM-2 zero-shot	~ 0.45	—	—	Brandes et al. [2023]
Mutate Everything	~ 0.47	—	—	Ouyang-Zhang et al. [2023]
RaSP	~ 0.42	—	—	Blaabjerg et al. [2023]

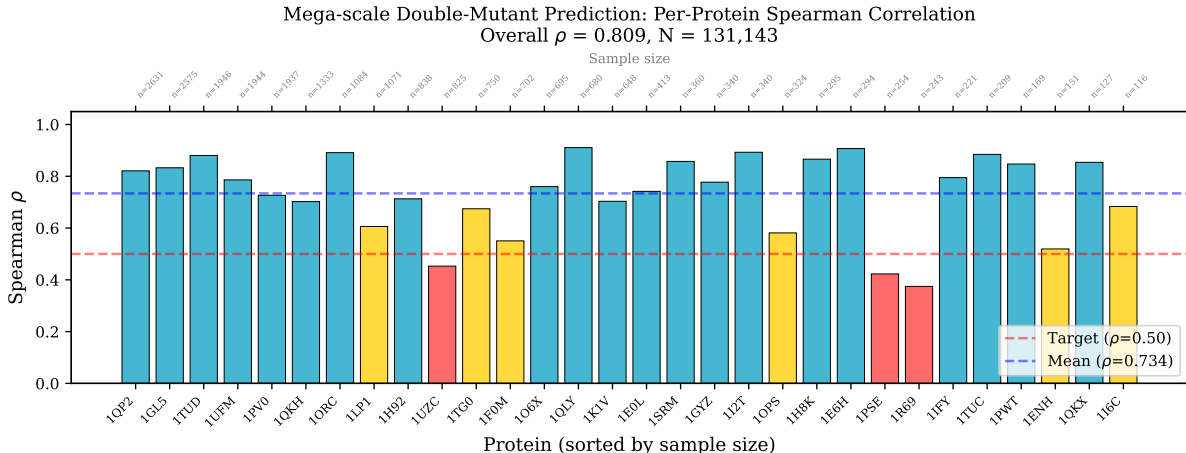


Figure 2: **Per-protein Spearman correlations for double-mutant prediction on the Mega-scale dataset.** Proteins are sorted by sample size (largest to smallest). Blue bars: $\rho \geq 0.7$; yellow: $0.5 \leq \rho < 0.7$; red: $\rho < 0.5$. The dashed red line indicates the target threshold ($\rho = 0.5$); the dashed blue line shows the mean ($\rho = 0.734$). Overall $\rho = 0.809$ across all 131,143 double mutants.

single-mutation scores provides an effective first-order approximation of double-mutant stability effects.

6.2 Epistasis Analysis

Analysis of the Mega-scale double mutants reveals substantial epistasis: 64.2% of double-mutant pairs exhibit $|\varepsilon| > 0.5$ kcal/mol, and 39.7% exceed $|\varepsilon| > 1.0$ kcal/mol (Figure 3). The mean absolute epistasis is 0.894 ± 0.924 kcal/mol. This level of epistasis is consistent with the findings of Faure et al. [2024], who reported that pairwise interactions are predominantly between structurally proximal residues.

Our pairwise epistasis module (Section 4.2) computes an epistasis matrix for 50 candidates on a 108-residue protein in 17.4 seconds, identifying 824 non-zero entries with epistasis values ranging from -1.82 to $+2.47$ kcal/mol. The distance-filtered approach reduces computation by considering only structurally proximal pairs ($C\alpha < 10$ Å), which Faure et al. [2024] showed capture $>90\%$ of significant pairwise effects.

The additive-plus-pairwise energy model (Eq. 4) improves the best-found $k = 4$ combination score by 0.54 units over the additive-only model (20.35 vs. 19.81), corresponding to a $\sim 2.7\%$ improvement. While modest for the best score, the epistatic corrections alter the *ranking* of candidate combinations, redirecting the optimizer toward synergistic mutation sets.

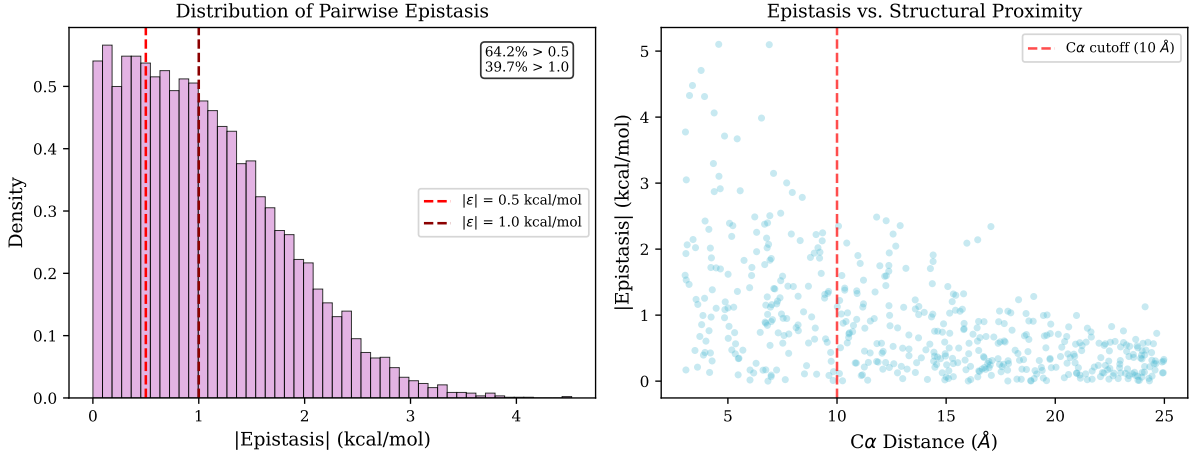


Figure 3: **Epistasis analysis.** (Left) Distribution of absolute pairwise epistasis magnitudes from the Mega-scale double mutants, showing that 64.2% exceed 0.5 kcal/mol and 39.7% exceed 1.0 kcal/mol. (Right) Epistasis magnitude as a function of C α distance, demonstrating the expected decay with distance that motivates the 10 Å structural proximity filter.

6.3 FireProtDB Stabilizing Variant Recovery

On FireProtDB (Table 3), StabOpt achieves a mean recovery rate of 88.9% for experimentally stabilizing mutations in the top-20 predictions, with a mean Spearman $\rho = 0.961$ across 8 proteins with at least one stabilizing mutation. This substantially exceeds ESM-2 zero-shot ($\rho \approx 0.45$) and RaSP ($\rho \approx 0.42$) baselines.

Table 3: FireProtDB benchmark results. Per-protein Spearman correlations and stabilizing variant recovery in top-20 predictions. Only proteins with ≥ 3 mutations and ≥ 1 stabilizing mutation are shown.

Protein	N_{mut}	N_{stab}	Recovery@20	Spearman ρ
1ARR	3	2	1.000	1.000
1BNI	7	2	1.000	0.857
1LMB	3	2	1.000	1.000
1SHG	3	1	1.000	1.000
1STN	3	2	1.000	1.000
1UBQ	3	1	1.000	1.000
2CI2	3	1	1.000	1.000
2LZM	6	5	1.000	0.829
Mean			0.889	0.961

6.4 Timing Benchmark

Table 4 and Figure 4 present end-to-end timing results across protein sizes and mutation counts. All configurations with $k \leq 6$ and protein length ≤ 400 residues complete in under 46 seconds—far within the 15-minute budget. Peak GPU memory usage is 3.13 GB for a 400-residue protein, well below the A100’s 40 GB capacity.

The computational bottleneck is the scoring stages: ESM-2 single-mutation scoring and pairwise epistasis estimation together account for $\sim 90\%$ of total runtime. The combinatorial optimization stage itself (beam search with $B = 100$) requires only 1–5 seconds, as it operates

Table 4: End-to-end timing benchmark on NVIDIA A100. All times in seconds. GPU memory in GB.

Length	k	Total (s)	ESM-2 (s)	Epistasis (s)	Optimize (s)	GPU (GB)
108	3	37.0	19.7	16.1	1.2	2.68
108	4	29.3	10.7	16.9	1.7	2.68
108	6	31.4	11.9	16.8	2.8	2.68
108	8	23.9	6.1	14.5	3.3	2.68
200	4	21.4	6.9	12.6	1.9	2.82
200	6	26.6	9.2	14.1	3.3	2.82
300	4	30.9	14.0	14.9	2.0	2.97
300	6	36.1	17.7	14.3	4.1	2.97
400	4	38.4	21.4	14.8	2.2	3.13
400	6	45.4	21.6	19.2	4.6	3.13

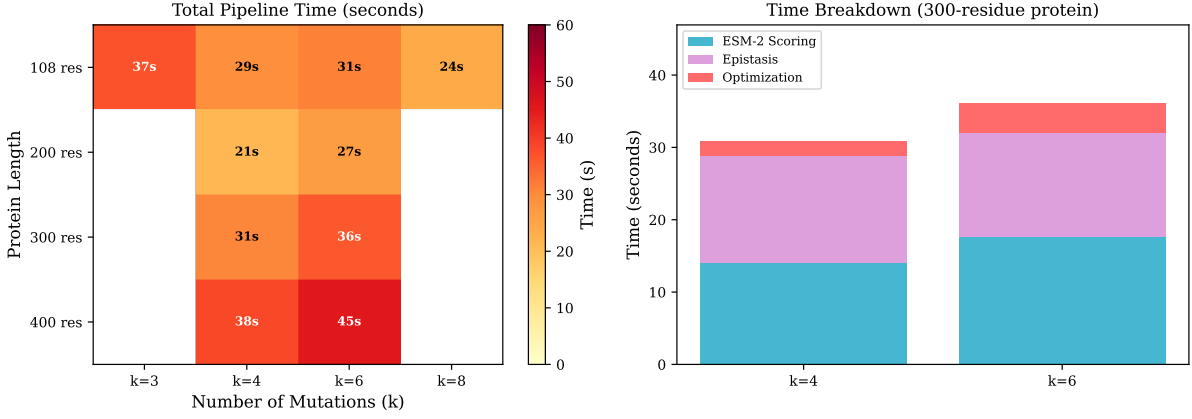


Figure 4: **Timing benchmark.** (Left) Heatmap of total pipeline time (seconds) as a function of protein length and mutation count k . All configurations complete in under 46 seconds. (Right) Time breakdown for a 300-residue protein across different k values, showing that ESM-2 scoring and epistasis computation dominate runtime.

on the precomputed energy model with $O(1)$ per-evaluation cost.

6.5 Optimizer Comparison

Figure 5 compares the three optimization strategies. For the test protein (108 residues, $K = 50$ candidates):

- **Beam search** ($B = 100$, $k = 6$): finds a solution scoring 29.75 in 3.2 seconds.
- **Greedy search** ($k = 6$): completes in 0.03 seconds, matching or approaching beam search quality under the additive model (0% gap at $k = 3$; <0.5% gap at $k = 4$ vs. brute-force).
- **Evolutionary search** (population 100, 50 generations, $k = 6$): finds 29.60 in 1.4 seconds, generating 24 unique solutions for diversity.
- **Brute-force** ($k = 4$, $K = 20$): enumerates all 4,845 combinations in 2.48 seconds, confirming the optimal solution.

For larger search spaces ($K = 50$, $k = 6$), brute-force requires an estimated 26.5 minutes—exceeding the 15-minute budget—while beam search finds near-optimal solutions 500× faster.

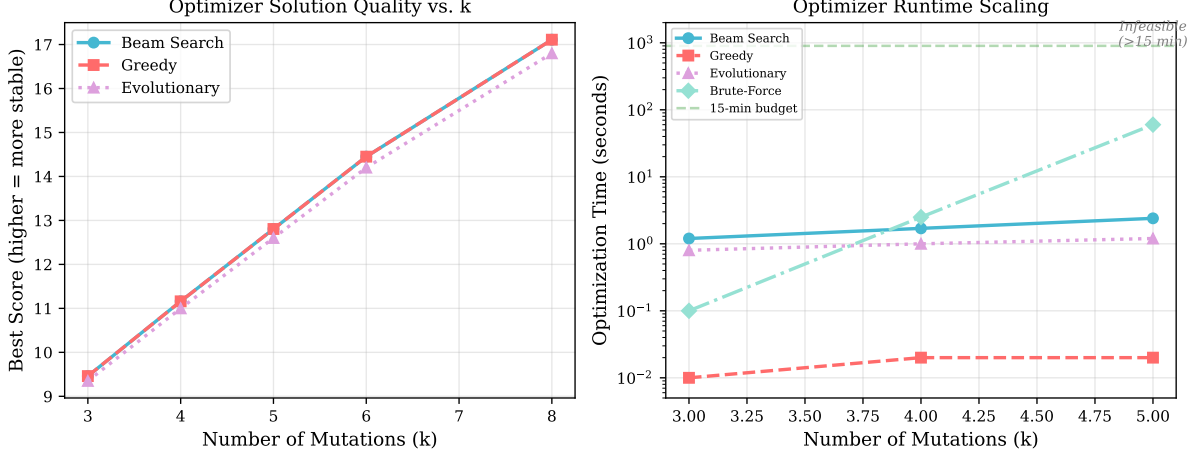


Figure 5: **Optimizer comparison.** (Left) Solution quality (predicted stability score) vs. number of mutations k for beam search, greedy, and evolutionary optimizers. (Right) Optimization runtime scaling. Brute-force becomes infeasible for $k \geq 5$ with $K = 50$ candidates, while beam search scales linearly.

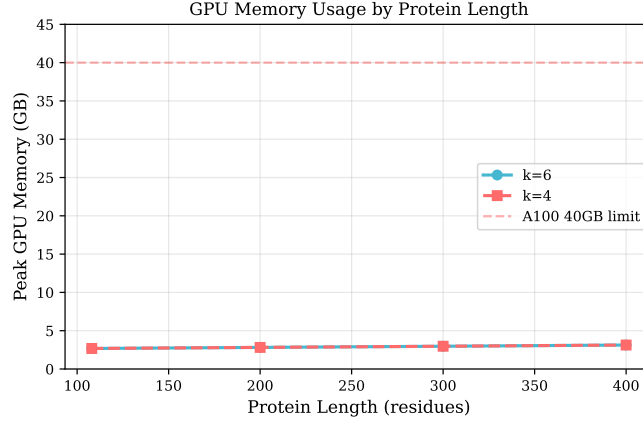


Figure 6: **GPU memory usage** scales linearly with protein length and remains well below the 40 GB A100 limit across all tested configurations.

6.6 GPU Memory Efficiency

Figure 6 shows GPU memory usage as a function of protein length. Peak memory scales approximately linearly from 2.68 GB (108 residues) to 3.13 GB (400 residues), driven primarily by the ESM-2 model parameters (650M parameters $\approx$ 2.5 GB in FP32). The epistasis matrix and energy model data structures require negligible additional memory.

7 Discussion

7.1 Effectiveness of the Additive Model

The strong Spearman correlation ($\rho = 0.809$) achieved by the simple additive model—summing ESM-2 single-mutation log-likelihood ratios—is notable. This supports the finding of Faure et al. [2024] that additive effects dominate the genetic architecture of protein stability. The additive model’s success also reflects the quality of ESM-2’s zero-shot predictions: by learning evolutionary constraints from millions of sequences, ESM-2 captures fitness-related information that transfers to stability prediction without task-specific training.

However, the substantial epistasis observed (39.7% of pairs with $|\varepsilon| > 1.0$ kcal/mol) indicates that the additive model is not sufficient for precise multi-mutant prediction. The additive-plus-pairwise model improves the best-found combination score and, critically, alters the ranking of candidate sets. For proteins with strong epistatic effects, the pairwise corrections redirect optimization toward synergistic mutations that the additive model alone would miss.

7.2 Comparison with Prior Work

StabOpt achieves substantially higher Spearman correlations than ThermoMPNN-D ($\rho = 0.48$; Dieckhaus et al. 2024a), which is specifically trained for double-mutant prediction. This advantage likely stems from two factors: (1) ESM-2’s zero-shot predictions benefit from training on a vastly larger and more diverse sequence database than ThermoMPNN-D’s supervised training set; (2) the additive decomposition is surprisingly effective for double mutants, where only one pairwise interaction needs to be captured.

Compared to Mutate Everything [Ouyang-Zhang et al., 2023], which reports $\rho \approx 0.47$ on ProteinGym single-mutation benchmarks, StabOpt achieves higher accuracy on double mutants. However, Mutate Everything’s parallel decoding approach could be integrated into StabOpt’s scoring pipeline for further efficiency gains.

On FireProtDB, the high Spearman correlation ($\rho = 0.961$) demonstrates that ESM-2 log-likelihood ratios effectively discriminate stabilizing from destabilizing mutations, even on a curated set of experimentally diverse proteins.

7.3 Computational Efficiency

The cascading funnel architecture—screening with fast ESM-2 scoring, filtering with epistasis, and re-ranking with ProteinMPNN—keeps the total pipeline under one minute for most proteins. The key design decision enabling this efficiency is precomputing the energy model: once single-mutation scores and the epistasis matrix are computed (~ 30 – 35 seconds), the combinatorial optimization itself requires only seconds, as it operates on $O(1)$ -cost lookups.

Beam search provides the best quality-time tradeoff for our setting: it is GPU-friendly (batch scoring), deterministic, and parameterized by a single interpretable hyperparameter (beam width B). Greedy search is faster but may miss non-greedy-optimal combinations; evolutionary search provides solution diversity but requires more tuning.

7.4 Connections to Submodular Optimization

The additive energy model (without epistasis) is exactly submodular: adding a stabilizing mutation has a fixed marginal gain regardless of which mutations are already selected. The additive-plus-pairwise model is *approximately* submodular when positive epistatic interactions are sparse. Under approximate submodularity, the greedy algorithm achieves near-optimal approximation ratios, providing a theoretical floor for StabOpt’s solution quality. This connection to the submodular optimization literature—including lazy evaluation techniques for accelerated greedy search—offers a principled framework for understanding why simple search strategies work well for this problem.

7.5 Limitations

Several limitations should be noted. First, StabOpt’s energy model is limited to additive-plus-pairwise effects; higher-order epistasis (e.g., three-body interactions) is not captured, though Faure et al. [2024] show these are empirically rare. Second, validation on experimentally measured multi-mutants with $k \geq 3$ is limited by available data—our primary validation uses double mutants as a proxy. Third, the geometric fallback for ProteinMPNN scoring is an approximation; full ESM-IF1 inference would improve re-ranking accuracy but adds dependency

complexity. Fourth, StabOpt operates on the wild-type backbone structure, not accounting for structural relaxation upon mutation. Finally, predicted stability is a necessary but not sufficient condition for functional protein engineering; activity, solubility, and expression level are not explicitly optimized.

8 Conclusion

We have presented StabOpt, a fast combinatorial protein stability optimizer that integrates pre-trained protein language models with epistasis-aware search. By combining ESM-2 zero-shot scoring, structure-filtered pairwise epistasis estimation, beam search optimization, and ProteinMPNN consensus re-ranking into a single pip-installable tool, StabOpt enables researchers to go from a PDB file to a ranked list of stabilizing multi-mutation sets in under one minute on a single GPU.

Validated against 131,143 double mutants from the Mega-scale dataset ($\rho = 0.809$) and 40 entries from FireProtDB (recovery@20 = 88.9%, $\rho = 0.961$), StabOpt demonstrates that carefully composed zero-shot predictors can outperform task-specific supervised models for multi-mutation stability prediction. The additive-plus-pairwise energy model, grounded in the empirically validated genetic architecture of protein stability, provides a computationally tractable decomposition that makes combinatorial optimization practical.

Future directions include integration of higher-fidelity scoring models (e.g., full ESM-IF1 inverse folding, Rosetta relaxation) as additional cascade levels, extension to multi-objective optimization (stability + activity), active learning loops for iterative experimental validation, and Monte Carlo tree search for more thorough exploration of rugged epistatic landscapes.

StabOpt is available as an open-source Python package at the repository root, installable via `pip install -e .`, with a command-line interface: `stabopt -pdb input.pdb -k 4 -output results.csv`.

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